$f_3 \cdot f_2(S_2)\}$

$\left(\frac{S_2}{2}\right)^2\}$

$\left(\frac{S_3-y_3}{2}\right)^2\}$

$\frac{-y_3)^2}{4}\Big] = 0$

$(S_2-y_3)+(S_3-y_3)\Big] = 0$

Solution of LPP by New Modified backward DPP Procedure

(2) Max $Z = 4x_1 + 14x_2$

s.t.o.c,

$2x_1 + 7x_2 \le 21$

$7x_1 + 2x_2 \le 21$

and $x_1, x_2 \ge 0$

Soln

using backward procedure

$f_2(B_{12}, B_{22}) = \text{Max} \{14x_2\}$

$0 \le 7x_2 \le B_{12}$

$= 14 \text{ Max} \{x_2\}$

$0 \le x_2 \le \frac{B_{12}}{7}$

$0 \le x_2 \le \frac{B_{22}}{2}$

$= 14 \min \left\{\frac{B_{12}}{7}, \frac{B_{22}}{2}\right\}$

because Max $\{x_2\}$ satisfies

$0 \le x_2 \le \frac{B_{12}}{7}$, $0 \le x_2 \le \frac{B_{22}}{2}$ is

The minimum of $\frac{B_{12}}{7}, \frac{B_{22}}{2}$

$f_1(B_{11}, B_{21}) \ge \max \{4x_1 + f_2$

$(B_{11} - 2x_1, B_{12} - 7x_1)\}$

$= \max \{4x_1 + 14 \min\left(\frac{B_{11}-2x_1}{7}, \frac{B_{21}-7x_1}{2}\right)\}$

$0 \le x_1 \le B_{11}/2$

$0 \le x_1 \le B_{21}/7$

at first stage

$B_{11} = 21$, $B_{21} = 21$

$f_1(B_{11}, B_{21}) = f_1(21, 21)$

$= \max_{0 \le x_1 \le \frac{21}{2},\ 0 \le x_1 \le 8} \left\{ 4x_1 + 14 \min\left(\frac{21-2x_1}{7}, \frac{21-7x_1}{2} \right) \right\}$

$= \max_{0 \le x_1 \le 3} \left\{ 4x_1 + 14 \min\left(\frac{21-2x_1}{7}, \frac{21-7x_1}{2} \right) \right\} \rightarrow (2)$

now $\min\left(\frac{21-2x_1}{7}, \frac{21-7x_1}{2} \right) = \begin{cases} \frac{21-2x_1}{7}, & \text{if } 0 \le x_1 \le \frac{7}{3} \\ \frac{21-7x_1}{2}, & \text{if } \frac{7}{3} \le x_1 \le 3 \end{cases}$

Therefore from (2)

$f_1(B_{11}, B_{21}) = \max_{0 \le x_1 \le 3} \left\{ 4x_1 + 14 \min\left(\frac{21-2x_1}{7}, \frac{21-7x_1}{2} \right) \right\}$

$f_1(B_{11}, B_{21}) = \text{Max} \begin{cases} 42, & \text{if } 0 \le x_1 \le 7/3 \\ 147 - 45x, & \text{if } \frac{7}{3} \le x_1 \le 3 \end{cases}$

rough

$4x_1 + 42 - 4x$

$4x_1 + 147 - 49x$

Let $x_1 = \lambda$

$x_2 = \text{Min} \left\{ \frac{21-2x_1}{7}, \frac{21-7x_1}{2} \right\}$

$= \text{Min} \left\{ \frac{21-2\lambda}{7}, \frac{21-7\lambda}{2} \right\} = \frac{21-2\lambda}{2}$ where $0 \le \lambda \le \frac{7}{3}$ and

max $z = 42$.

① Solve the following LPP using Dynamic programming principles.

Max $Z = 2x_1 + 5x_2$

s.to $2x_1 + x_2 \le 43$

$2x_2 \le 46$

and $x_1, x_2 \ge 0$

Soln The problem consist of 2 decision variables and 2 constraints.

Hence the problem has 2 stages and 2 state variable

Let (B_{1j}, B_{2j}) be the state of the system at stage 1 $\S$

$f_j(B_{1j}, B_{2j})$ be the optimal value of the objective function for stage 1 & 2. Using backward computational procedure, we have

$$f_2(B_{12}, B_{22}) = \max_{\substack{0 \le x_2 \le B_{12} \\ 0 \le 2x_2 \le B_{22}}} \{5x_2\}$$

$$= 5 \max_{\substack{0 \le x_2 \le B_{12} \\ 0 \le x_2 \le \frac{B_{22}}{2}}} \{x_2\}$$

Since Max $\{x_2\}$ which satisfies $0 \le x_2 \le B_{12}$, $0 \le x_2 \le \frac{B_{22}}{2}$ is the min $\left\{B_{12}, \frac{B_{22}}{2}\right\}$

$$x_2^* = \max\{x_2\} = \min\left\{B_{12}, \frac{B_{22}}{2}\right\}$$

$$f_2(B_{12}, B_{22}) = 5 \min\left\{B_{12}, \frac{B_{22}}{2}\right\} \rightarrow (1)$$

$$f_1(B_{11}, B_{21}) = \max_{0 \le 2x_1 \le B_{11}} \{2x_1 + f_2(B_{11} - 2x_1, B_{21} - 0)\}$$

$$f_2(B_{11} - 2x_1, B_{21}) = 5 \min\left\{B_{11} - 2x_1, \frac{B_{21}}{2}\right\}$$

$$f_1(B_{11}, B_{21}) = \max_{0 \le x_1 \le \frac{B_{11}}{2}} \left\{2x_1 + 5\left(\min B_{11} - 2x_1, \frac{B_{21}}{2}\right)\right\}$$

Rough
$43 - 2x_1 = 23$
$43 - 23 = 2x_1$
$20 = 2x_1$
$x_1 = 10$

from constraint $B_{11} = 43$, $B_{21} = 46$

$$f_1(B_{11}, B_{21}) = \max_{0 \le x_1 \le \frac{B_{11}}{2}} \left\{2x_1 + 5 \min\left(43 - 2x_1, \frac{46}{2}\right)\right\}$$

$$\min(43 - 2x_1, 23) = \begin{cases} 23, & \text{if } 0 \le x_1 \le 10 \\ 43 - 2x_1 & \text{if } 10 \le x_1 \le 2 \end{cases}$$

$$f_1(B_{11}, B_2) = \max_{0 \le x_1 \le \frac{43}{2}} \left\{2x_1 + 5 \begin{pmatrix} 23 & \text{if } 0 \le x_1 \le 10 \\ 43 - 2x_1 & 10 \le x_1 \le 23 \end{pmatrix}\right\}$$

$$\max_{0 \le x_1 \le \frac{43}{2}} \begin{cases} 2x_1 + 115 & \text{if } 0 \le x_1 \le 10 \\ 2x_1 + 215 - 10x_1, & \text{if } 10 \le x_1 \le 23 \end{cases}$$

135
245

$$x_2 = \min \left\{ B_{11} - 2x_1, \frac{B_{21}}{2} \right\}$$

$$= \min \left\{ 43 - 2(10), \frac{46}{2} \right\}$$

$$\boxed{x_2 = 23}$$

$$f = \max \left\{ 2(10) + 115 \quad | \quad 215 - 8(10) \right\}$$

$$= 135$$